

Nitya Patel

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SUMMARY

Backend software engineer with 2+ years of professional experience building distributed, event-driven systems and platform tooling in Python. Track record of owning systems end-to-end — from design to production — across high-volume trading infrastructure, applied ML, and API-driven applications.

EDUCATION

IIIT Vadodara <i>B.Tech in Computer Science GPA: 8.61/10.0</i>	Gandhinagar, Gujarat 2020 – 2024
Shri Agrasen Vidyalaya <i>Higher Secondary Certificate Score: 84.6%</i>	Indore, MP 2019

EXPERIENCE

Graviton Research Capital <i>Software Developer – Tech Tools</i> - Develop and maintain daily automations and exchange-specific workflows across 4 exchanges, ensuring reliable execution with consistent uptime. - Build and own the <code>config_converter</code> module, enabling zero-manual-intervention runs at 1000s of executions daily. - Design a usage-based borrow allocation algorithm used across ~30 strategies. - Create internal tooling to eliminate repetitive tasks and improve team productivity.	Gandhinagar / Gurugram June 2024 – Present
VizExperts <i>Software Development Intern</i> - Designed and built an AI-based live drift correction module from scratch using Python, OpenCV, and PyTorch, targeting iPad AR workflows without multi-camera hardware support. - Achieved a ~10–15× reduction in drift accumulation rate compared to baseline. - Assisted with bug resolution in an AR/VR application built on Unreal Engine using C++ and Blueprint scripting.	Gurugram January 2024 – May 2024
MathWorks <i>EDG Intern</i> Added support for FMU 3.0 in the FMU Import Block component in Simulink.	Bangalore May 2023 – July 2023
iHub-Data, IIIT Hyderabad <i>Machine Learning Intern</i> - Contributed to a Dataset Foundation project to create a platform for ML/AI researchers. - Collaborated with a team of 10 to design the project architecture and establish an efficient delivery pipeline. - Engineered a high-throughput, multithreaded download manager using Python's <code>threading</code> module for concurrent dataset retrieval up to 100 GBs.	Hyderabad May 2022 – July 2022

PROJECTS

Algorithmic Trading Framework <i>Python, asyncio, REST APIs, Parquet</i> - Architected an event-driven trading infrastructure with a central EventBus — fully decoupled components enable new exchanges or strategies as single-file additions without touching existing code. - Implemented a live CoinDCX exchange adapter consuming REST APIs for OHLCV candle streaming, with a BaseExchange interface enabling multi-exchange support without architectural changes.
Ornithological Sound Identifier <i>Python, PyTorch</i> Implemented a multilayered neural network that identifies birds from soundscapes.
Todoist Command Line Utility <i>Python</i> - Developed a command line utility for Todoist using the Todoist REST API. - Handled API authentication, paginated resource fetching, and local state caching for a responsive offline-tolerant CLI experience.

TECHNICAL SKILLS

Languages: Python, C, C++, Java, JavaScript, Matlab, SQL, Bash
Tools & Frameworks: Git, Django, ReactJS, REST APIs, $\LaTeX$, Firebase, PyTorch
Operating Systems: Windows, Linux

ACHIEVEMENTS

Giftabled Hackathon: 2nd rank among 100 participants — created a facial recognition app embedded on an attachable pin, focused on helping visually impaired people.
CTF: 3rd rank among 100 participants — Capture the Flag; solved sequential puzzles involving cryptography, computer networks, web technology, and maths.
Tech Hunt: 1st rank among 150 participants — non-sequential puzzle competition; highest number of puzzles solved wins.